
AI Powered Digital Gold Investment Assistant

1. Introduction

Digital financial platforms today often use AI assistants to guide users in making investment decisions. These assistants help users understand different financial products and sometimes recommend suitable investment options based on the user's query. One such feature is the AI-based recommendation system used in modern fintech applications.

The objective of this project was to simulate a simplified version of such a system using APIs. In this project, an AI-like assistant interacts with a user, identifies whether the question is related to gold investment, and then suggests investing in digital gold. If the user decides to proceed, the system processes the purchase and records the transaction.

The project demonstrates how a basic AI workflow can be connected with backend services to create an automated investment recommendation and purchase system.

2. Objective of the Project

The main objective of this project is to design a simple backend system that performs the following tasks:

- Accept a user query related to investment.
- Identify whether the query is about gold investment.
- Provide a short response about gold investment.
- Suggest digital gold purchase.
- Process the purchase request through an API.
- Store the transaction details in a database.

The system is designed to show how AI recommendations can be integrated with a financial transaction workflow.

3. Technologies Used

The project was implemented using the following tools and technologies:

- **Python** – Used as the main programming language for backend development.

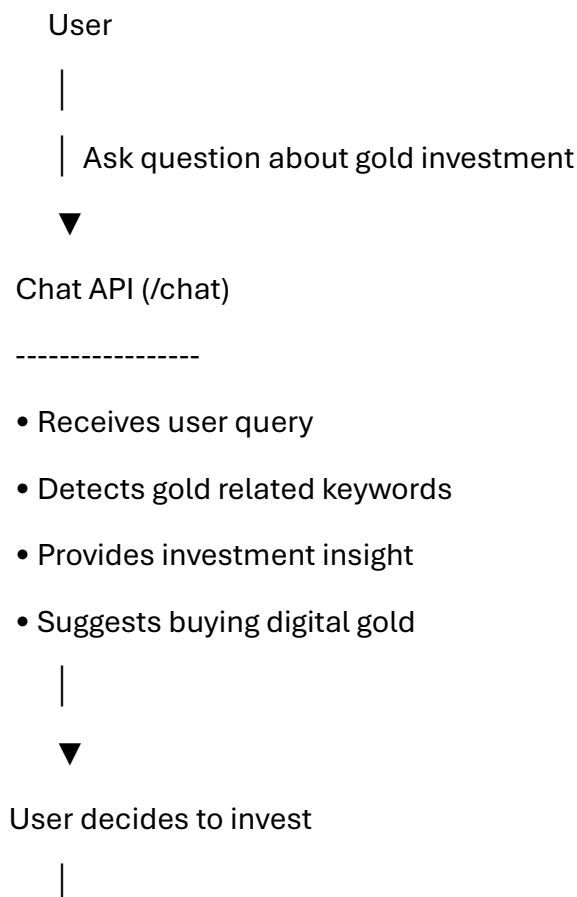
- **FastAPI** – Used to build REST APIs that handle user requests and responses.
- **Uvicorn** – Used as the server to run the FastAPI application.
- **SQLite** – Used as a lightweight database to store digital gold purchase transactions.
- **Google Colab** – Used as the development environment for writing and testing the code.
- **ngrok** – Used to expose the local API server to the internet so that it can be accessed externally.

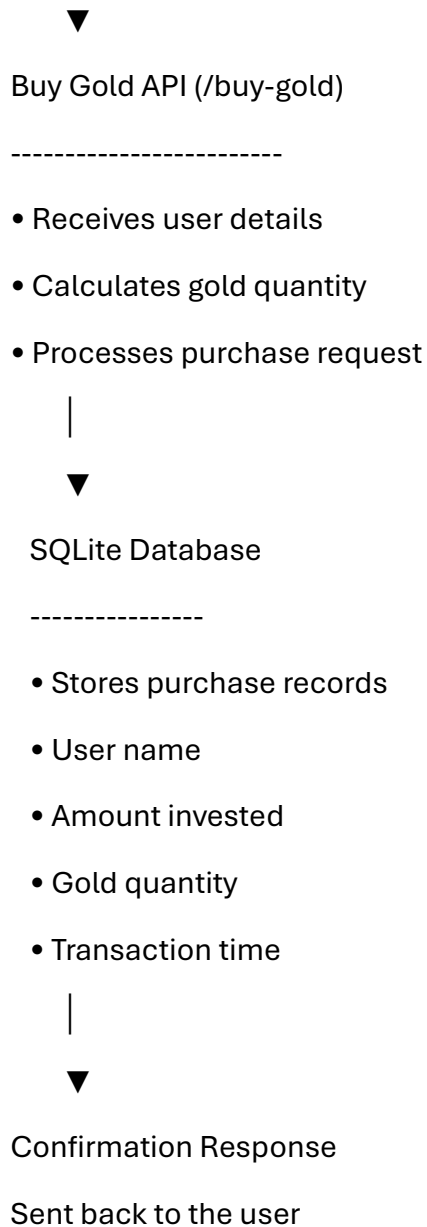
These tools were selected because they are lightweight, easy to use, and suitable for demonstrating API-based workflows.

4. System Architecture

The system is divided into two main components: the Chat API and the Buy Gold API. The Chat API handles the interaction with the user, while the Buy Gold API handles the purchase process.

Architecture Diagram





The architecture shows how the user interacts with the system through APIs and how the system processes and stores the transaction data.

5. System Workflow

The workflow of the system can be described in the following steps:

1. The user sends a message asking about gold investment.
2. The Chat API receives the message and analyzes it.
3. If the query contains keywords related to gold investment, the system responds with an investment insight.
4. The system also suggests purchasing digital gold through the platform.

5. If the user chooses to proceed, the Buy Gold API is called.
6. The API calculates the amount of gold that can be purchased based on the investment amount.
7. The purchase information is stored in the database.
8. The system returns a confirmation message indicating that the digital gold purchase was successful.

This workflow demonstrates how an AI-based assistant can guide a user from asking a question to completing a transaction.

6. Database Design

A SQLite database named **gold.db** is used to store transaction data. The database contains a table called **gold_purchases** which records all purchase transactions.

Table Structure

Column	Description
id	Unique transaction ID
user_name	Name of the user
gold_amount	Quantity of gold purchased
price	Amount invested
date	Date and time of the purchase

Every time the Buy Gold API is called, a new record is inserted into this table. This helps maintain a history of all transactions.

7. API Endpoints

Chat API

Endpoint:

POST /chat

This API receives the user's message and checks whether it is related to gold investment. If it detects a relevant query, it returns a response suggesting digital gold investment.

Example request:

```
{  
  "message": "Should I invest in gold?"  
}
```

Example response:

```
{  
  "answer": "Gold is considered a safe investment and protects against inflation.",  
  "nudge": "You can invest in digital gold through our platform."  
}
```

Buy Gold API

Endpoint:

POST /buy-gold

This API processes the digital gold purchase request. It receives the user's name and investment amount, calculates the gold quantity, and stores the transaction in the database.

Example request:

```
{  
  "user_name": "Sushant",  
  "amount_in_inr": 5000  
}
```

Example response:

```
{  
  "message": "Digital gold purchased successfully",  
  "gold_grams": 0.833,  
  "amount_paid": 5000  
}
```

8. Deployment

The project was developed and tested in Google Colab. Since the server runs locally inside the notebook environment, ngrok was used to create a public URL that exposes the API to the internet.

This allowed the APIs to be accessed and tested through a browser using the automatically generated API documentation interface.

9. Challenges Faced

During the development of the project, a few challenges were encountered:

- Running the FastAPI server in the Google Colab environment required handling asynchronous event loop issues.
- Setting up ngrok authentication and generating a working public URL.
- Debugging API errors caused by incorrect JSON request formats.

These challenges were resolved through debugging and adjustments in the server setup.

10. Future Improvements

The current system uses keyword detection to identify gold-related queries. In a real-world application, this could be improved by integrating a proper AI language model that understands user queries more accurately.

Other improvements could include:

- Integration of real-time gold price APIs.
 - Adding a user interface for chat interaction.
 - Implementing authentication for users.
 - Providing a dashboard for viewing transaction history.
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11. Conclusion

This project demonstrates how a simple AI-driven recommendation system can be integrated with backend APIs to simulate a digital investment workflow. The system successfully detects gold-related queries, suggests digital gold investment, processes the purchase request, and stores the transaction data.

Although the project is a simplified implementation, it effectively shows how AI assistants can be combined with financial services to create interactive investment platforms.